



Exam Questions

Space I

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Exercises

Match each pioneer with their field of study:

- | | |
|----------------|------------------------------------|
| 1. Newton | a. Rocket Propulsion |
| 2. Kepler | b. Force, gravitational attraction |
| 3. Copernicus | c. Celestial mechanics |
| 4. Hohmann | d. Solar system |
| 5. Tsiolkovsky | e. Interplanetary flight |

Exercises

Who performed the first manned flight?

a. John Glenn, b. Neil Armstrong, c. Yuri Gagarin, d. Aleksei Sputnik

Who performed the first US manned flight?

a. John Glenn, b. Neil Armstrong, c. Buzz Aldrin, d. Virgil Grissom

Exercises

What is the name of the largest rocket ever?

a. Mercury-Redstone 1, b. Luna, c. Vostok-L, d. Saturn V

Who was the first man on the moon?

a. Buzz Aldrin, b. Yuri Gagarin, c. Neil Armstrong, John Glenn

When was the first dutch satellite (ANS) launched?

a. 1974, b. 1971, c. 1969, d. 1983

Exercises

What is the altitude boundary that defines space?

- a. 50 km
- b. 80 km
- c. 100 km
- d. 120 km

Which of the following is NOT a challenge faced in space?

- a. No oxygen
- b. Strong atmospheric drag
- c. Radiation exposure
- d. Extreme temperature changes

Exercises

Which company is NOT a major player in modern commercial spaceflight?

- a. SpaceX
- b. Boeing
- c. Blue Origin
- d. General Motors

What is the primary advantage of reusable rockets?

- a. Increased thrust
- b. Reduction in launch costs
- c. Faster launch times
- d. Improved accuracy

Exercises

Which propulsion method does NOT rely on chemical reactions?

- a. Solid rocket engines
- b. Liquid rocket engines
- c. Ion propulsion
- d. Hybrid rocket engines

What does the Tsiolkovsky Rocket Equation calculate?

- a. Maximum altitude of a rocket
- b. Final velocity of a rocket
- c. Amount of thrust required for launch
- d. The temperature of a rocket engine

Exercises

Why can't airplanes be used to reach space?

- a. They do not generate enough speed
- b. Their engines require atmospheric oxygen
- c. They experience too much drag
- d. All of the above

What is the main function of an upper stage in a rocket?

- a. Protect the payload
- b. Provide final velocity for orbital insertion
- c. Increase fuel efficiency
- d. Control the spacecraft during re-entry

Exercises

A rocket engine expels gases at an exhaust velocity of 3,200 m/s with a mass flow rate of 250 kg/s. What is the thrust produced by the engine?

- a) 500,000 N
- b) 800,000 N
- c) 1,000,000 N
- d) 1,250,000 N

(*Use the thrust equation)

Exercises

A rocket has an initial mass of 500,000 kg and a final mass of 100,000 kg. The effective exhaust velocity is 3,000 m/s. What is the final velocity of the rocket according to Tsiolkovsky's Rocket Equation?

- a) 3,230 m/s
- b) 4,828 m/s
- c) 9,234 m/s
- d) 12,154 m/s

Exercises

A spacecraft needs to achieve a velocity change of 9,500 m/s. Its initial mass is 300,000 kg, and its final mass after burning fuel is 75,000 kg. What must be the minimum exhaust velocity for this mission?

- a) 3,523 m/s
- b) 4,276 m/s
- c) 5,539 m/s
- d) 6,852 m/s

Exercises

A rocket engine generates 1,200,000 N of thrust with an exhaust velocity of 3,600 m/s. What is the required mass flow rate of the expelled gases?

- a) 208 kg/s
- b) 252 kg/s
- c) 333 kg/s
- d) 357 kg/s

Exercises

A spacecraft needs to reach a final velocity of 8,000 m/s. It has an effective exhaust velocity of 3,100 m/s. If the dry mass is 20,000 kg, what should the fuel mass be?

- a) 244,113 kg
- b) 255,753 kg
- c) 265,948 kg
- d) 275,127 kg

Exercises

A rocket on Earth has a thrust of 3,500,000 N and a total mass of 250,000 kg. What is its thrust-to-weight ratio (TWR)?

- a) 1.2
- b) 1.4
- c) 1.6
- d) 1.8

Exercises

You are designing a rocket, definitely not for military purposes.

You can choose between 2 engines.

Both engines have a burn time of 89 seconds, and an exhaust velocity of 1800 m/s.

Engine 1 has a mass flow of 120 kg/s, whilst engine 2 is heavier but has a mass flow of 180 kg/s.

From calculations you know that your minimum thrust must be 300 kN.

Which rocket do you use?

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What is the mass of the propellant you have to take on board?

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If the final velocity is 3000 m/s, what is the dry mass of the rocket?

Answers

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3. Copernicus - Solar system
4. Hohmann - Interplanetary flight
5. Tsiolkovsky - Rocket Propulsion

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Rocket 2

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16,020 kg

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3730.36 kg